Azure DevOps

1. **Introduction**
    a. What is DevOps
    b. A brief history about DevOps
    c. Goals of DevOps
    d. DevOps Vs Traditional Silos
2. **DevOps Concepts and Practices**
    a. Build Automation
    b. Continuous Integration
    c. Continuous Delivery and Continuous Deployment
    d. Infrastructure as a Code
    e. Configuration Management
    f. Orchestration
    g. Monitoring
    h. Microservices
3. **DevOps Tools**
    a. Tools for Build Automation and Continuous Integration
    b. Tools for Continuous Integration
    c. Tools for Virtualization and Containerization
    d. Tools for Monitoring
    e. Tools for Orchestration
4. **DevOps and the Cloud**
    a. DevOps and Azure
5. **Continuous Planning**
    a. Introduction to Agile
    b. Various agile practices with Azure DevOps
    c. Azure Project Wiki
6. **Continuous Versioning with Azure Repos**
    a. Source Control
    b. Exploring Azure Repos
    c. Configuring branches
    d. Branching Strategies
    e. Pull Request Workflow
    f. Using Git tags to Organise a repository
    g. Exploring Repository Permissions
    h. Azure Repos Demo and Git Workflow
    i. Repository Sharing with Submodules
    j. Connecting to GitHub Using Azure Active Directory

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THE STUDENTS WE'VE RECENTLY PLACED



D. Mahesh

Rsspa it solutions

Job Role: Front-end Web Developer



D. Abhishek

Arun Kapital Networks

Role: Network Engineer L-1



K.Pavn Manohar

nipun

Job Role: Associate Network engineer



Racha Lokesh

JNET Technologies

Job Role:Trainee .Net developer



Bandi ramya

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Job Role: Associate Network engineer



P. Shirisha

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Job Role: Associate Network engineer



D.Bala Vinay

Arun Kapital Networks

Job Role: Network Engineer



D. Madhusudan

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Job Role: Associate Network engineer



P. Venkannababu

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Role: Associate Network Engineer



M.Ashok Reddy

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Job Role: Associate Network engineer



Velde. Srivastava

TATA CONSULTANCY SERVICES

Job Role: Assistant System Engineer Trainee



Mohammed Zubair

STPL

Job Role: Network Engineer L1



G. Rithwini Sagar

Arun Kapital Networks



V. Venkateswarlu

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A.Pradeep Chand

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P. Venkata Harish

SEC COMMUNICATIONS

Job Role: Technical Engineer



J.Sudhakar

Arun Kapital Networks



CH. Ashwith

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G.Poorna Satya Murthy

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M. Srikanth

nipun



P.Rakesh

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Job Role: Associate Network engineer



D. Rajesh Babu

Arun Kapital Networks

Job Role: Network Engineer



K.Balaraju

nipun

Job Role: Associate Network engineer



Ansari

Web Werks DATA CENTERS | IRON MOUNTAIN DATA CENTERS

Job Role: Network Engineer L1



Santosh Kumar

STPL

Job Role: Network Engineer L1



K. Srinivas Rao

SEC COMMUNICATIONS
INTEGRATING CUSTOMERS WITH TECHNOLOGY

Job Role: Network Engineer L1



B.SUDHEENDRA REDDY

LAURUS Labs
Knowledge...Innovation...Excellence

Role: ASSOCIATE IT ENGINEER

Hear from Our Students!

Hemabindhu Jonnalagadda
1 review • 0 photos

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I have recently joined in aimmxt instutite for the data analytics course the teaching is outstanding.They make us to understand each and every topic in easy way.We can freely ask any type of doubts about the topic.I am happy to my decision to join in this instutite.

Shaik Farida
1 review • 0 photos

★★★★★

I recently joined Aimnxt for the Data Analyst course, I'm happy with my decision. The teaching quality is excellent. If there are any complex concepts they breakdown it and make us easy to understand every topic. We can freely ask any kind of doubts about the topic. And also staff is friendly and supportive, always ready to assist us. I'm sure it's a great place to learn and grow.



Madina Charan
Local Guide • 14 reviews • 17 photos

★★★★★

Yeah I got a good experience and friendly teaching. AimNxt is hear to encourage new talents

Rithwini Sagar
2 reviews • 0 photos

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Providing good placement and trainings with good knowledge and skills.



sai sudhakar
2 reviews • 0 photos

★★★★★

The best edutech providing quality of training 👌 👌 👌



venky vr
1 review • 0 photos

★★★★★

Those who are wants to settle career in networking side this is the best coaching institute I have ever seen for freshers and upskilling students also.



Shankar
2 reviews • 0 photos

★★★★★

The instructors were incredibly knowledgeable and passionate about the subjects they were teaching. They went above and beyond to ensure that each participant understood the material and were always available for any questions or clarifications.

K

Kaya Charan
1 review • 0 photos

★★★★★

I would highly recommend AIMNXT Technology Institute to anyone looking to enhance their skills or gain specific knowledge in the tech industry.the commitment of the instructors make it an excellent choice for professional development.

M

M Priya
2 reviews • 0 photos

★★★★★

The best edutech company which provides quality trainings with best faculty and placement assistance to the students. Excellent infrastructure and committed faculty..



Nulu Veera Venkata Ratna Vineetha
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★★★★★

It was a great experience for me with Aimnxt technologies.. They are providing good training I recommend every one to join in Aimnxt technologies

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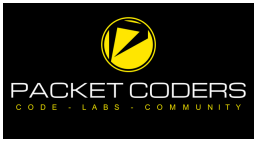


























k. Exploring Code Reviews
l. Exploring Static Code Analysis
m. Using Pull Requests with Work Items

7. **Configuring Repositories**
    a. Using Git Tags to Organize Your Repository
    b. Handling Large Repositories
    c. Exploring Repository Permissions
    d. Removing Repository Data
    e. Recovering Repository Data
8. **Continuous Build with Azure Pipelines**
    a. Azure Pipelines
    b. Understanding Build Agents
    c. Self Hosted Build Agents vs Microsoft hosted Agents
    d. Agent Pools
    e. Parallel Jobs
    f. Classic Vs YAML pipelines
    g. Pipeline as a Code (YAML)
    h. Agent Demands
    i. Multi Agent Builds
    j. Using Build Trigger Rules
    k. Incorporating Multiple Builds
    l. Exploring Containerized Agents
    m. GitHub Integration with Azure DevOps Pipelines
    n. Integrate External Services with Azure Pipelines
    o. Visual Studio Marketplace Demo
    p. Exploring Testing Strategies in Your Build
    q. Understanding Code Coverage
    r. Discovering Pipeline Health Monitoring
    s. Improving Pipeline Performance and Cost Efficiency
    t. Incorporating Variable Groups
9. **Continuous Integration with Azure Pipelines**
    a. Create a Release Pipeline
    b. Build and Release Tasks
    c. Exploring Stages, Dependencies, and Conditions
    d. Discovering Azure App Configuration
    e. Implementing Release Gates
    f. Implementing Deployment Slots Releases
    g. Implementing Load Balancer and Traffic Manager Releases
    h. Understanding Feature Toggles
10. **Analyzing Metrics**
    a. What Is Site Reliability Engineering (SRE)?
    b. Metric Charts and Dashboards

c. Application Health Checks
d. System Load and Failure Conditions
e. Baseline Metrics
f. Discovering Application Insights Smart Detection and Dynamic Thresholds
g. Deciding Which Dependencies to Set Alerts On

11. Implementing a package management Strategy

a. Discovering Package Management Tools
b. Exploring Azure Artifacts
c. Creating a Versioning Strategy for Artifacts

12. Designing and Implementing Infrastructure as Code (IaC)

a. Exploring Configuration Management
b. Introducing PowerShell Desired State Configuration (DSC)
c. Implementing PowerShell Desired State Configuration (DSC) for Application Infrastructure
d. Discovering Azure Policy
e. Exploring Azure Policy Guest

13. Developing Deployment Scripts and Templates

a. Understanding Deployment Solution Options
b. Exploring Infrastructure as Code: ARM vs. Terraform
c. Discovering Azure Bicep
d. Exploring Infrastructure as Code: PowerShell vs. CLI
e. Linting ARM Templates
f. Deploying a Database
g. Understanding SQL Data Movement
h. Introduction to Visual Studio App Center
i. Exploring CDN and IoT Deployments
j. Understanding Azure Stack and Sovereign Cloud Deployments

14. Managing Sensitive information in Automation

a. Exploring Azure AD Privileged Identity Management (PIM)
b. Exploring Azure AD Conditional Access
c. Implementing Multi-Factor Authentication
d. Working with Service Principals
e. Working with Managed Identities
f. Using Service Connections
g. Incorporating Vaults
h. Exploring Azure Pipelines Secrets
i. Discovering GitHub Secrets

15. Continuous code Analysis

a. Static Code Analysis
b. Dynamic code Analysis
c. Understanding Dependency Scanning
d. Exploring Container Dependency Scanning

e. Incorporating Security into Your Pipelines
f. Scanning for Compliance with tools like WhiteSource Bolt and SonarQube

16. **Configuring Monitoring for a DevOps Environment**
    a. Discovering Logs in Azure
    b. Introducing Azure Monitor Logs
    c. Controlling Who Has Access to Your Logs
    d. Discovering Crash Analytics
    e. Introducing Distributed Tracing
    f. Understanding User Analytics with Application Insights and App Center
    g. Exploring Infrastructure Performance Indicators
    h. Configure Alerts for Pipeline Events with Slack and Teams
    i. Exploring Monitoring Feedback Loops
17. **Communication and Collaboration**
    a. Exploring Custom Dashboards
    b. Incorporating Work Items with Deployments
    c. Using GitHub with Azure Boards
18. **Generating DevOps Process Documentation**
    a. Onboarding New Employees
    b. Exploring Azure DevOps Dependency Tracking

Azure Admin

1. **Cloud Concepts**
    a. Introduction
    b. The language of Cloud Computing
    c. The Economy of Cloud Computing
    d. Cloud Service Models
    e. Azure MarketPlace
    f. Cloud Architect Models
2. **Azure Architecture**
    a. Regions and Availability Zones
    b. Resource Groups and ARM
    c. Demo: Creating Azure Resources
    d. Azure Resource Manager
    e. Azure Portal and Azure Cloud Shell
    f. Azure CLI and Powershell
    g. Using ARM Templates
3. **Azure Networking**

a. Virtual Network
b. Demo: Create a Virtual Network
c. Load Balancer
d. VPN Gateway
e. Application Gateway
f. Content Delivery Network
g. ExpressRoute
h. NSGs
i. Azure DNS
j. Azure Firewall
k. Service Endpoints and Private EndPoints
l. Intersite Connectivity
m. Vnet Peering

4. **Azure Storage**
   a. Storage Accounts
   b. Blob
   c. Disk
   d. File
   e. Archive
   f. Blob Object Replication
   g. Blob Life Cycle Management
   h. Azure File Sync
5. **Azure Compute Options**
   a. Virtual Machines
   b. Demo: Create a VM
   c. Scale sets
   d. App Services
   e. Azure Container Instances
   f. Azure Kubernetes Service
   g. Azure Virtual Desktop
   h. Azure Functions
   i. Demo: Azure App Services
6. **Identity and RBAC**
   a. Azure Active Directory
   b. Users, Tenants, Groups
   c. Multi-Factor Authentication
   d. Single Sign-On
   e. SSPR
   f. AD Device Management
   g. Roles in Azure
   h. Assigning Access to Resources
   i. Custom Roles

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7. **Monitoring**
    a. Azure Monitor
    b. Setting up Alerts and Actions
    c. Azure Monitor Logs
    d. Monitor Insights
    e. Application Insights
    f. Network Watcher
8. **Backup and Recovery**
    a. Disaster Recovery
    b. Azure Backup
    c. Azure Site Recovery
    d. Backup Reports
9. **Database options in Azure**
    a. Cosmos DB
    b. Azure SQL
    c. Azure Database for MySQL
    d. Azure Database for PostgreSQL
    e. Database Migration Services
10. **Azure Security**
    a. Defence in Depth
    b. Securing Network Connectivity
    c. Public and Private Endpoints
    d. Microsoft Defender for Cloud
    e. Key Vault
    f. Azure Information Protection
    g. Microsoft Defender for Identity
    h. Azure Sentinel
    i. Azure Dedicated Hosts
    j. Demo: Azure Key Vault
11. **Azure - Privacy, Compliance and Trust**
    a. Governance
    b. Azure Monitor
    c. Monitoring Tools
    d. Azure Service Health
    e. Azure Arc
12. **Azure Pricing, Governance and Compliance**
    a. Managing Subscriptions
    b. Cost Management
    c. Pricing Factor
    d. Demo: Using the pricing Calculator
    e. Azure Policies
    f. Cloud Governance Strategy

13. **Azure Support**
    a. Plans
    b. Tickets
    c. Channels
    d. SLAs
    e. Service Life Cycle

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g. Managing Container Resources
h. Monitoring Container Health with Probes
i. Building Self-Healing Pods with Restart Policies
j. Creating Multi-Container Pods
k. Introducing Init Containers

5. **Advanced Pod Allocation**
   a. Exploring K8s Scheduling
   b. Assigning a Kubernetes Pod to a Specific Node
   c. Using DaemonSets
   d. Using DaemonSets in Kubernetes
   e. Using Static Pods in Kubernetes
6. **Deployments**
   a. K8s Deployments Overview
   b. Scaling Applications with Deployments
   c. Managing Rolling Updates with Deployments
   d. Deploying with Blue/Green and Canary Strategies
   e. Installing Helm
   f. Using Helm
   g. Deploying Packaged Kubernetes Apps with Helm
7. **Networking**
   a. K8s Networking Architectural Overview
   b. CNI Plugins Overview
   c. Understanding K8s DNS
   d. Using NetworkPolicies
8. **Services**
   a. K8s Services Overview
   b. Using K8s Services
   c. Exposing Kubernetes Pods Using Services
   d. Using Kubernetes Services With DNS
   e. Managing Access from Outside with K8s Ingress
   f. Using Kubernetes Ingress
   g. Controlling Network Access with NetworkPolicies - Part 2
   h. Exposing Applications with Ingress
9. **Storage**
   a. K8s Storage Overview
   b. Using K8s Volumes
   c. Exploring K8s Persistent Volumes
   d. Using K8s Persistent Volumes
10. **Application Observability and Maintenance**
    a. Understanding the API Deprecation Policy
    b. Implementing Probes and Health Checks
    c. Implementing Health Checks in a Kubernetes Application

d. Monitoring Kubernetes Applications
e. Accessing Container Logs
f. Debugging in Kubernetes

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Ansible

1. **Understanding Core Components of Ansible**
    a. Getting Started with Ansible
    b. A Brief Tour of the Ansible Configuration File
    c. Ad-Hoc Ansible Commands
2. **Inventory Essentials and Inventory Variables**
    a. Variables and Inventories
    b. Using YAML in Inventories
    c. Dynamic Inventories
    d. Working with Ansible Inventories
3. **Create Ansible Plays and Playbooks**
    a. Introduction to Playbooks and Common Modules
    b. Create Playbooks to Configure Systems to a Specified State
    c. Basic Playbook Syntax Demonstration
    d. Use Variables to Retrieve the Results of Running Commands
    e. Use Conditionals to Control Play Execution Part 1
    f. Use Conditionals to Control Play Execution Part 2
    g. Configure Error Handling
    h. Demo: Error Handling – Ignore Errors
    i. Demo: Error Handling – Block Groups
    j. Selectively Run Specific Tasks In Playbooks Using Tags
4. **Create and Use Templates to Create Customized Configuration Files**
    a. Using Ansible Templates Lecture
    b. Demo: Using Ansible Templates
5. **Work with Ansible Variables and Facts**
    a. Ansible Variables Lecture
    b. Ansible Variables - Magic Variables and Jinja Filters
    c. Ansible Variables - Variable Files
    d. Ansible Facts Lecture
    e. Working with Ansible Facts
6. **Create and Work with Roles**
    a. Working with Ansible Roles Lecture
    b. Applying In-Line Roles and Role Dependencies

c. Working with Ansible Roles
d. Download roles from an Ansible Galaxy
e. Ansible Galaxy

7. **Managing Parallelism**
    a. Parallelism in Ansible
8. **Use Ansible Vault in Playbooks to Protect Sensitive Data**
    a. The Ansible-Vault Command
    b. Using Vaults in Playbooks
    c. Working with Confidential Data in Ansible
9. **Install Ansible Tower and Use it to Manage Systems**
    a. Introduction to Ansible Tower
    b. Installing Ansible Tower
    c. Use Documentation to Look Up Specific Information About Ansible Modules and Commands
    d. Finding Documentation
10. **Install and Configure Control Node and Ansible Nodes**
    a. Shell Scripts to Run Ad-Hoc Commands
    b. Firewall Rules
    c. Archiving
    d. Scheduled Tasks: Cron
    e. Scheduled Tasks: `at`
11. **Security**
    a. Ad-Hoc Ansible shell scripts
    b. Ansible Archiving
    c. Ansible Cron
    d. Ansible Scheduling with AT
    e. Ansible Security

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